Dhruv Nirmal

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PROFESSIONAL SUMMARY

I build and maintain statistical and machine learning models that turn large, often messy internal and external datasets into decisions a business can act on, from time series forecasting to classification, and I care as much about validating and documenting a model properly as about building it in the first place. Across procurement, pricing, fresh produce inventory planning and government transport analytics, including a classification pipeline built for a client's roughly A$12B spend book, I've carried models from a first prototype through feature engineering, cross validation and hyperparameter tuning to a documented, repeatable process a wider team can rely on. I've also helped turn manual, one off reporting into automated workflows built directly on top of analytics and modelling outputs, and I keep clear documentation so whoever maintains a model after me isn't starting from scratch. I'd bring that same mix of practical model building, rigorous validation and a habit of documenting and automating what I build to EA's Publishing Insights and Markets team.

KEY SKILLS

Statistical & ML Modelling: Python, forecasting (Prophet), classification (logistic regression, TF-IDF and embeddings), feature engineering, cross-validation, hyperparameter tuning, class-imbalance handling, clustering and segmentation fundamentals (postgraduate coursework), anomaly detection

Languages & Data: Python, SQL, pandas, scikit-learn, Snowflake

Visualisation & BI: Tableau, Power BI, dashboard design, trend and KPI reporting

Tools & Practices: Git and GitHub (version control), model and pipeline documentation, Databricks and PySpark (distributed data processing), AI/LLM assisted workflow automation (Claude, ChatGPT)

EXPERIENCE

Data Scientist, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

- Own quantitative analysis and pricing-decision dashboards end to end for a multi-country restaurant chain across Australia and New Zealand, tracking price movement and surfacing cost-leakage trends across products, venues and suppliers for senior stakeholders including the Chief Procurement Officer; helped the client find close to 30% savings in one of their largest spend categories over a year.
- Built the agentic AI workflow that automatically assembles structured client progress reports from analytics and data science outputs, rolled out company-wide across roughly 25-30 client accounts and saving the analyst team about 12-15 hours a week.
- Coordinate and document data-validation conventions across a global, multi-region client pipeline spanning 13 sub-regions, standardising file submissions and automated checks so reporting stays consistent as the pipeline scales.
- Applied statistical and ML modelling across a range of commercial environments within the role, from procurement and pricing to fresh-produce inventory planning, adapting the same feature-engineering and validation approach to each new dataset.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a distributed anomaly-detection pipeline in Databricks using PySpark on large-scale government procurement data, engineering scalable transformation layers and surfacing irregular spend patterns; improved detection accuracy by about 20%.
- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders, presenting the model architecture and outcomes to a cross-functional audience.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO
Jan 2021 to Jul 2021 | India

- Analysed blast-wave signal data using statistical modelling and Butterworth filtering to predict noise levels, improving prediction accuracy by about 20% and refining defence-equipment designs.

PROJECTS

Inventory Demand Forecasting Model (Python, Prophet)

- Built a time series forecasting model for a fresh-produce client to plan raw-material and chemical inventory three months ahead, incorporating external price drivers such as sea-freight trends; delivered forecasts within a 12.5-14% error margin, comfortably inside the client's accepted tolerance, through a dashboard for ongoing planning.

Client Spend Categorisation Classifier (Python, scikit-learn)

- Led a one-vs-all classification model (TF-IDF and embeddings, logistic regression) to categorise a client's roughly A$12B, five-year spend dataset, engineering features, tuning hyperparameters, handling class imbalance and validating with cross-validation; turned a categorisation cycle that used to take an account manager one to one and a half months into a single day's model run, and documented the pipeline for reuse across the wider client base.

Job Hunt OS, personal build (Python, Claude Code, Git/GitHub)

- Designed and built a file-based, multi-agent career-search system that tailors resumes to a job description, drafts cover letters, and produces daily and weekly reports, maintained in a version-controlled GitHub repository; a self-directed way to keep learning agentic AI tooling and good repository practice outside of client work.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

- Coursework: Statistics I and II, Machine Learning, Applied Forecasting, High Dimensional Analysis, Communicating with Data, Supply Chain Management.

Bachelor of Engineering, Thapar University | May 2017 to May 2021